

Optics

THE SCIENCE OF OPTICS EXPLAINS AND EXPLORES THE PROPERTIES AND BEHAVIOR OF LIGHT.

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Light is a type of electromagnetic radiation. It is carried by a stream of particles that can also behave like a wave.

Light sources

The Sun, lights, and TV screens all emit (send out) light—they are luminous. But most objects reflect and/or absorb light that bounces off them. Transparent materials, such as glass and water, let light pass right through them. They transmit light.

Features of light

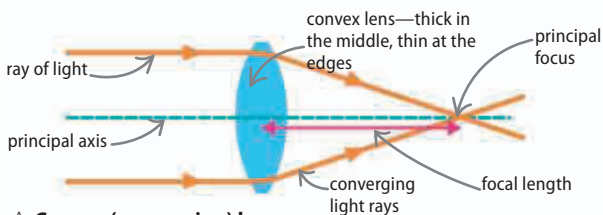
form of radiation	Light is a form of electromagnetic radiation (see pages 194–195). It radiates (spreads out) from its source.
light rays travel in straight lines	You can see this in the beams from lighthouses, flashlights, and lasers. Because light rays are straight, if an object blocks them you get a dark region of shadow.
transfers energy	Energy is needed to produce light. All materials gain energy when they absorb light—solar cells use the energy in sunlight to produce electricity.
stream of particles that can behave like a wave	Light is carried by a stream of particles, called photons, but in some situations this stream can also behave like a wave.
can travel through empty space	Electromagnetic waves do not need to travel through a medium (a material such as water or air). The light from the Sun and stars, for example, reaches us through empty space.
travels fast	Light is the fastest thing in the Universe. In a vacuum, such as space, its speed is exactly 299,792 km per second (roughly 186,282 miles per second).

△ Understanding light

The most important source of light on Earth is the Sun. Sunlight is produced by the energy generated deep in its core (see pages 232–233). In contrast, the Moon simply reflects the light of the Sun and shines much less brightly. The table above gives the main features of light.

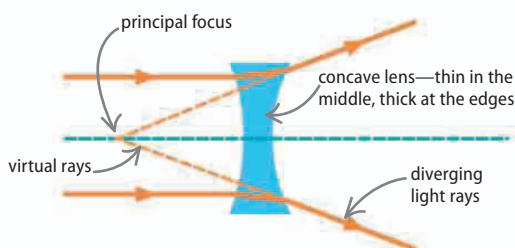
Lenses

A lens is a piece of transparent glass or plastic that uses refraction (see page 197) to change the directions of light rays. Lenses are used to focus light in glasses, cameras, and telescopes. There are two main types—convex and concave.



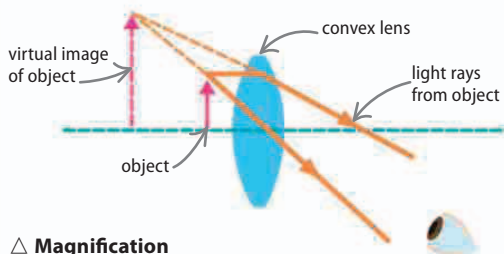
△ Convex (converging) lens

When rays pass through a convex lens they converge (bend inward) and meet at a point behind the lens, called the principal focus. The distance from the center of the lens to the principal focus is called the focal length.



△ Concave (diverging) lens

A concave lens makes light diverge (spread out). When parallel rays pass through a concave lens they spread out as if they came from a focal point in front of the lens, called the principal focus.



△ Magnification

If an object is placed between the center of a convex lens and the principal focus, the rays never converge. Instead they appear to come from a position behind the lens as a magnified image. It is a virtual image (see page 197).